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Review

RTO: An overview and assessment of current practice

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ABSTRACT

The practice of implementing real-time optimization (RTO) using a rigorous steady-state model, in conjunction with model predictive control (MPC), dates back to the late 1980s. Since then, numerous projects have been implemented in refinery and chemical plants, and RTO has received significant attention in the industrial and academic literature. This history affords us the opportunity to assess the impact and success of RTO technology in the process industries. We begin with a discussion of the role RTO serves in the hierarchy of control and optimization decision making in the plant, and outline the key steps of the RTO layer and the coordination with MPC. Where appropriate, we point out the different approaches that have been used in practice and discuss the success factors that directly relate to the success of RTO within an organization. We also discuss alternative approaches that have been used to alleviate some of the challenges associated with implementing RTO and which may be appropriate for those unwilling to commit to the traditional RTO approach. Lastly, we provide suggestions for improvement to motivate further research.

Contents

1. Introduction

In the mid to late 1980s, the confluence of several developments allowed for the first time, real-time optimization utilizing a rigorous steady-state model of the process:

- Model predictive control technology.
- Open equation modeling.

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- Computer processing capability (speed, memory, affordable cost).
- Large scale, sparse matrix SQP solvers.

Rigorous modeling implies the use of multi-component mass and energy balances, vapor–liquid equilibrium expressions, and reaction kinetics; however, some amount of empirical approaches may be required to describe some effects that are not easily modeled, such as hydraulic effects or reaction kinetics. While not all RTOs implemented over the past 20 or so years have utilized rigorous models, the majority have; therefore, we refer to this approach as “traditional RTO”. On the basis of publications and our knowledge of the industry, we estimate that, to date, there are 250–300 implementations of RTO utilizing commercially available rigorous flowsheeting packages based on open-equation modeling techniques. This total does not include RTO applications that have been developed in-house. A major contribution to the use of rigorous models was the advent of open equation modeling, which allowed engineers to focus on the model equations without worrying about convergence details, or how to structure and solve the model update step vs. the optimization step, as was the case with traditional, flow sheeting packages [15]. Computer processing capability and sparse matrix SQP solvers allowed large problems to be solved efficiently and reliably.

Optimal operation typically resides at the intersection of multiple constraints. As a result, MPC has proved necessary to move the plant to, and maintain operation at, the steady-state optimum between sucessive executions of the RTO [6]. The basic structure of RTO in cascade to MPC has become the standard approach for implementing steady-state optimization in plants that operate around nominal steady states.

In the past 20 years, there has been continued improvement in computer processing capability, which has allowed bigger models and faster convergence for both MPC and RTO. For example, 100,000–200,000 equation models for optimization (with 30–40 decision variables) are fairly common today and typically solve both parameter fitting and optimization cases in less than 40 min elapsed time on common desktop computers There have also been significant changes in the software. What was once custom code is now mature, supported products (see, e.g., [35]).

At the same time, there have been significant changes that have negatively impacted RTO applications. Significant staffing reductions have affected both the operation groups and the support groups responsible for maintaining MPC and RTO. Budgets have become tighter, and there is increased competition for the available investment capital (e.g., IT and environmental). Increasingly, global competitive pressures have brought about more frequent changes to plant operation – both in terms of equipment modifications and operating conditions. As a result, there are greater challenges to keeping MPC and RTO solutions performing well [7], and this has led to a mixed record of success in maintaining these applications.

Today, we find industry split in the acceptance of RTO. Some companies are convinced of the benefits and continue to invest in applications; other companies have concluded that RTO, as it is currently implemented, is not viable for them. In the following we consider RTO in more detail and discuss the challenges involved in applying RTO as well as the factors that increase the likelihood of success. We also highlight alternative RTO approaches that have been used, including simplifications that result in an easier to maintain solution. Lastly, we provide suggestions for improvement that we hope will motivate further discussion and research within and between the academic and industrial communities.

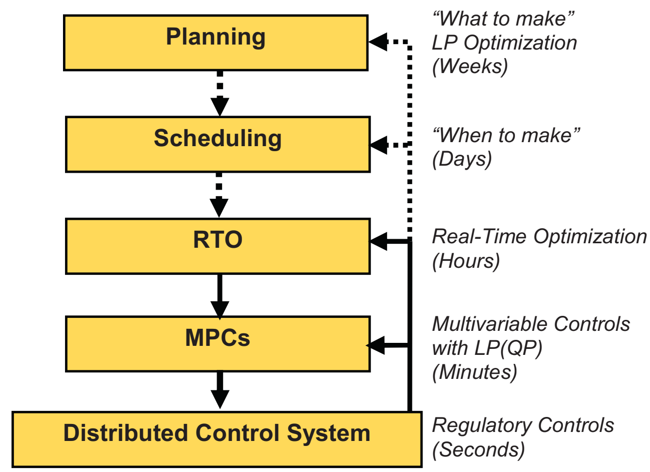

Fig. 1. Plant decision hierarchy.

2. Plant decision hierarchy

The general hierarchy for control and decision making in a plant is shown in Fig. 1. What is usually emphasized with this hierarchy is the successive refinement of time scales from top to bottom. In addition, there are also varying spatial scales, from plant-wide optimization at the top to regulatory control via single PID loops at the bottom.

This hierarchy, for which no claim of optimality is made, has evolved based on the decisions that must be made for an operating plant, with consideration to the available information uncertainties, and technology limitations. We note that a hierarchical approach is a common solution to managing complex systems and distributing decision making (see, e.g., [9]). This is a crucial differentiating factor that decomposes an otherwise unmanageable problem (i.e., an all-encompassing optimization in real time) into a cascade of interconnected solvable problems. Successful model building in a hierarchy includes determining what is relevant for the specific decisions that must be made and omitting the details that are not relevant [21]. What is generally lacking today is in the sharing of relevant model information, constraints and pricing across all levels of the hierarchy.

Planning is concerned with “what and how” based on economics and forecasts, and answers such questions as what feedstocks to purchase, which products to make, and how much of each product to make. In most all refineries and larger chemical plants, a linear program (LP) or successive LP (SLP) is used for planning and is based on an overall plant profit objective function.

Scheduling is concerned with “when”. Scheduling addresses the timing of actions and events necessary to execute the chosen plan, with the key consideration being feasibility. Scheduling deals with such issues as the timing of the deliveries of feeds, product liftings and operating mode changes, and avoiding storage problems (overflow or shortage). A range of tools are used across the industry for scheduling; from whiteboards and spreadsheets to tools involving simulation models, rules, and optimization [44].

The RTO layer and the levels below it execute in real-time, albeit at different intervals, and there is automatic, continual feedback from the process (indicated by the solid lines entering from the right). At the planning and scheduling levels, feedback and model updating is not automatic and is performed intermittently (indicated by dotted lines in the figure). Outputs from planning and scheduling are shown as dotted lines to signify that human transla-

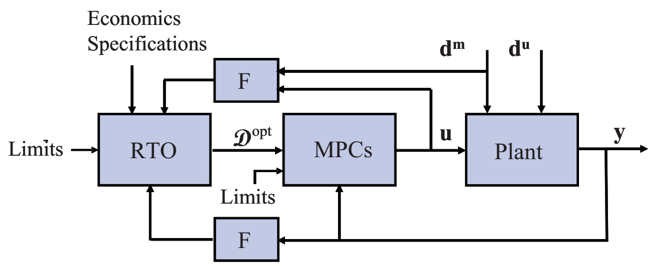

Fig. 2. Simplified block diagram. **F** denotes filtering/averaging; $\mathbf{d}^m$ represents measured disturbances variables and $\mathbf{d}^u$ unmeasured disturbance variables; **u** and **y** represent the plant input and output variables, respectively; D^{opt} represents the optimal decisions that are passed from RTO to MPC.

tion is required to determine and set certain objectives of the next level, which are normally not simply a one-to-one translation of objectives to targets and limits. Day-to-day logistical issues from scheduling can have a significant impact on plant operations, overriding decisions that are made at the RTO level and requiring target and limit changes at the MPC level.

RTO is concerned with implementing business decisions in real time based on a calibrated non-linear steady-state model, with model detail that the planning model does not have. RTO is implemented where economically justified and is typically formulated based on a profit function of the plant. RTO values are directly passed or translated to values or parameters of the MPC.

At the MPC level, there may be one or more MPCs, depending on the controller envelope(s). The MPC(s) provides the minute-to-minute dynamic control of the plant and provide some amount of optimization capability.

The lowest layer is the distributed control system (DCS), which is responsible for regulatory control (typically PID) and usually executes on a second time scale (sub-second to multi-second). The DCS layer is typically the main operator interface for monitoring and controlling the plant.

Each level in the hierarchy may impact different departments in the organization. Success of RTO requires support of the entire organization. If not planned for, RTO (more so than MPC) can create conflict within the organization and lead to failure of the RTO application. This is a key point that cannot be overemphasized.

3. RTO and MPC: functionality and integration

It is important to further understand the problems that RTO and MPC solve and how they are typically integrated. Fig. 2 is a simplified block diagram of the combined system. The RTO and MPC are not connected in a traditional cascade arrangement, but in an internal model control structure. Although a steady-state model is used for RTO, when connected to a (dynamic) plant, the resulting system is dynamic. RTO can thus be viewed as a single step, model-based controller with a future horizon (control and prediction) of just one point. Unmeasured disturbances affect both MPC and RTO and include such effects as feed stock variations ambient effects, and plant upsets (e.g., taking equipment in or out of service). It is helpful to think of RTO-specific disturbances, which are related to changes in economics, efficiencies, and equipment fouling or catalyst deactivation that occur at lower frequencies but accommodated in real time.

Because RTO relies on a static model, the plant must be sufficiently steady to reliably update the plant model. Steady-state detection methods typically incorporate statistics such as the mean, variance, or slope for selected signals over a moving window [42]. Comparisons are then made against thresholds or previous values. Steady-state is deemed to occur if the selected signals pass specified tests. A common approach is to simply consider bound checks on the standard deviation of selected signals. In this case, the design problem is one of determining the subset of signals to use in the detection, and normally include slower-responding signals such as temperatures and compositions. The detection may also be performed based on the energy content (as a function of frequency) compared to a reference [2]. Filters and averages of plant measurements (**F** in Fig. 2) are used to remove high frequency disturbances and suppress dynamic effects. The RTO scheduling mechanism is normally the detection of steady-state following the previous implementation of optimal RTO decisions.

The rate at which RTO can be executed depends on the frequency of unmeasured disturbances and the time required for MPC to move the process to a new steady state. Complicating the situation is that there is not a clear separation of time scales, i.e., open-loop response times are similar since both consider the same variables (e.g., compositions), and MPC must first reject unmeasured disturbances in those variables before a new steady state can be achieved.

Current industrial practice is to validate input data primarily via bound checks, rather than by statistical data reconciliation/gross error detection techniques. If a validity check fails, alternate measurements or estimates may be automatically substituted when available. The logic is designed based on historical data and engineering judgement, and configured using standard software tools. More complex validation logic is sometimes implemented with custom scripting. For critical variables, if no suitable alternative value is available for substitution, the execution of the optimizer is aborted until valid data is available.

The RTO model is updated (calibrated) at each cycle to minimize the deviation between model variables and plant measurements. Model parameters such as distillation efficiencies, heat transfer coefficients and unmeasured flows are typically adjusted to accomplish this objective. Typical industrial practice is discussed further in Section 5.

Once the model is updated, the optimization case is performed. Typically, the objective function is based on profit. If the case converges successfully, optimal decisions are passed to MPC. These decisions (D^{opt} in Fig. 2) may take the form of desired targets for controlled or manipulated variables, upper/lower variable limits, or possibly costs. The majority of industrial MPC applications continue to rely on linear models, empirically developed from a dedicated plant test [8]. Nonlinear MPC, which is an active research area, could theoretically allow RTO and MPC to be combined in a single layer (or allow one model to be used). However, except for certain processes such as polymers, which tend to be relatively small control problems, nonlinear MPC has, to date, not found significant application in the wider industry. Linear models (with linearizing transforms, where appropriate) have shown to be generally adequate for many control problems encountered in the process industries [17].

There are different design practices in the industry today pertaining to the sizing (i.e., controller envelope) of the MPC(s). Some practitioners prefer designing multiple MPCs on the basis of individual process equipment (such as distillation column and reactor). Others prefer a design with a larger envelope to encompass plant constraints that can limit plant throughput.

There are various ways to accommodate RTO setpoints for controlled and manipulated variables in MPC. One way is to include an additional term in the MPC objective function to penalize deviations from RTO setpoints. Another formulation is via a separate target selection layer to determine the best, feasible setpoints for the MPC [33,47]. Here we assume the target selection is an LP (or QP) optimizer, executing at the same frequency as the MPC, which is common in the industry. An advantage of this two-stage implementation over the single stage formulation is that it can better respond to disturbances and maintain feasibility between RTO executions [47]. The typical LP formulation takes the following form:

$$\begin{aligned}&\min_{\mathbf{y}^{TG},\mathbf{u}^{TG}} \mathbf{c}_y^T\mathbf{y}^{TG} + \mathbf{c}_u^T\mathbf{u}^{TG}\\ &\text{s.t.}\\ &\mathbf{y}^{TG} = \mathbf{G}_{ss}\mathbf{u}^{TG} + \mathbf{b}\\ &\mathbf{y}^{\min} \le \mathbf{y}^{TG} \le \mathbf{y}^{\max}\\ &\mathbf{u}^{\min} \le \mathbf{u}^{TG} \le \mathbf{u}^{\max}\end{aligned} \tag{1}$$

where $\mathbf{c}_y$ and $\mathbf{c}_u$ are the costs associated with the outputs and inputs (controller manipulated variables), respectively; $\mathbf{y}^{TG}$ and $\mathbf{u}^{TG}$ are the optimal targets for the outputs and inputs, respectively; superscripts min and max refer to the constraint limits for the inputs and outputs; $\mathbf{G}_{ss}$ is the steady-state gain matrix obtained from the MPC dynamic model; and $\mathbf{b}$ is the steady-state model bias term, which is updated based on current output measurements and predictions from the dynamic model. Constraint priorities are handled either through the addition of soft penalty terms [47] or by solving a sequence of LPs [27]. Desired RTO setpoints can be introduced into the LP(QP) MPC by various means. Some packages include previsions to accommodate external setpoints. Another technique is to create additional controlled variables in the MPC to accommodate RTO setpoints (for either inputs or outputs). Minimum and maximum limits are set equal to create setpoints, which then become constraints in Eq. (1). Relatively low weights or priorities are used on these setpoints to avoid potential conflict with higher priority specifications and constraints.

The MPC regulator is typically posed as

$$\begin{aligned}&\min_{\mathbf{u}}(\mathbf{y}^{TG} - \hat{\mathbf{y}}_f)^T\mathbf{Q}(\mathbf{y}^{TG} - \hat{\mathbf{y}}_f) + \Delta\mathbf{u}^T\mathbf{R}\Delta\mathbf{u}\\ &\text{s.t.}\\ &\hat{\mathbf{y}}_f = \mathbf{g}(\mathbf{u})\end{aligned} \tag{2}$$

where $\mathbf{g}$ is the (linear) MPC dynamic model, linking future inputs, $\mathbf{u}$ to predicted outputs, $\hat{\mathbf{y}}_f$; $\mathbf{Q}$ is the output weighting matrix and $\mathbf{R}$ is the move suppression matrix, both of which are tuning parameters. Several of the commercial packages force the dynamic solution to equal the optimal steady-state solution ($\mathbf{y}^{TG}$, $\mathbf{u}^{TG}$) at the end of the prediction and control horizons.

Since both the RTO and MPC have their own objective functions, there is the potential for conflict, which can arise due to model differences – nonlinear in the RTO versus linear in the MPC – or from mismatch in the economics between the objective functions, if degrees of freedom exist in the MPC.

4. RTO justification

Both MPC and RTO can provide significant benefits. With MPC, the majority of the benefits come from the target selection function via the embedded LP(QP). It is important that RTO benefits be compared to those than can be obtained from MPC alone [17] by analyzing the specific opportunities being targeted. If optimal operation lies at constraints that can be consistently determined by MPC, RTO is not required. However, even if the optimum always lies at constraints, it may be that the constraint set can change and only the RTO is able to find the optimal set. In essence, RTO accounts for tradeoffs that MPC cannot, due to plant nonlinearities, or by considering a larger envelope. RTO benefits can be in the range of 0–50% (typical rule of thumb) of MPC benefits. Following implementation of RTO and MPC, it is usually difficult to determine the benefit contribution of each. An experimental approach is to turn off MPC and RTO for a brief period of time and then carefully measure the differences as each is turned on. This approach can be both expensive and inaccurate. Benefits are lost while RTO and MPC are turned off and it often takes a long period of time for the process to return to its "natural" state, leading to an underestimation of benefits, if the testing is done over a short period of time. From a technical point of view, it is relatively easy to use the MPC and RTO models to estimate the benefits using offline case studies, but such an approach may not satisfy management. Most audits are usually performed on the combined MPC–RTO system.

Some practitioners use RTO to provide the coordination function for individual MPCs (e.g., within an ethylene plant or fluid catalytic cracker (FCCU). This may reflect design preferences (i.e., smaller vs. larger MPCs – cf. Section 3), but may lead to different conclusions regarding the potential need and benefit of RTO for some situations. If the control problem is approximately linear such that the true optimum can reliably be tracked, an alternative (and less expensive solution) to RTO is a coordinator type of MPC which solves a common LP(QP). In this case, the overall model is constructed from the models of the individual MPCs. The resulting solution is passed to the regulators of the individual MPCs (see, e.g., [26,34]).

The benefits of MPC are usually of a "primary" type that follow directly from accepted operational objectives. Economic objectives that are directional (i.e., minimum or maximum constraint pushing) are easily accomplished with MPC. Examples include minimizing duty subject to specification limits and maximizing feed rate subject to plant capacity constraints.

The benefits or RTO tend to be "secondary" and therefore less intuitive and often more controversial. These are situations that cannot be handled with directional economics Candidate areas include:

- Conversion vs. throughput trade-offs.
- Combining reactors and downstream separations.
- Multiple units competing for limited or shared resource.
- Large and potentially variable product price differentials.
- Units with large recycle.

The key requirement for RTO is variability of the optimal operating point caused by changes in operating conditions, different feedstocks, variable constraints and major changes in economics (product and feedstock prices). If variability is minimal, the optimum policy can be determined offline on an infrequent basis, thereby avoiding the trouble and expense of closing the loop on the RTO system. In larger plants, seasonal and day-to-night variations may be sufficient to justify RTO.

Additional benefits may be realized if the RTO model is used offline for such tasks as:

- "What if" runs.
- Improvement studies.
- Updating of other models (e.g., planning).

Offline benefits result from using the calibrated online model since the RTO model is continually updated to the plant. There are also benefits from monitoring RTO model parameters such as equipment efficiencies and heat transfer coefficients as a means of optimally scheduling maintenance activities.

5. RTO current practice

In this section we provide technical details of the key issues involved in developing and deploying RTO.

5.1. RTO scope

The typical RTO scope tends to cover an entire unit. For example, in ethylene the scope is typically the entire plant (starting with the cracking furnaces and including at least the C2 and C3 Splitters). In oil refining the following are typical scopes: Crude Units (including preheat/crude column/vacuum column and associated distillation columns, FCCU (including reactor and downstream distillation columns), or Hydrocracker (including reactors, separation and distillation columns).

5.2. Model rigor and detail

In Section 1, we began the discussion by defining what we mean by rigorous models, including detailed component mass and energy balances, rigorous tray by tray separation calculations and detailed molecular kinetic reaction schemes. Implied also is that all equipment within the model scope be represented in the model at this level of detail. However, in practice, not everything must be modeled in maximum detail and rigor to achieve the desired economic objectives. An overly complicated model may be harder to match to the plant and harder to maintain, yet contribute little additional value over a more simplified model. Based on recent discussions with technology suppliers and end users, there is more emphasis today on value creation and ease of maintenance. For example, a recent RTO project for a FCCU rigorously modeled the reactor/regenerator and main fractionator, but chose a short-cut model for the light ends distillation columns.

Less rigorous modeling approaches include shortcut distillation models and reaction schemes that group kinetically similar molecules into kinetic "lumps". In addition, it is a design decision whether to model all process equipment explicitly or combine equipment into a single representative model (e.g., combine several heat exchangers into a single representative heat exchanger models with characteristics of the group). The key consideration is to model the process in enough detail and rigor to adequately reflect the key economic tradeoffs and operating constraints so that the overall model results in a valid economic optimum.

Another consideration for the modeling effort is what Gallun et al. [16] refer to as rational RTO implementation. A plant model may be representative of actual operation, but still be wrong from an operational and/or larger economic context. Consider an optimization problem that contains parallel, identical heaters. Due to small differences in the models (resulting from slight changes in efficiency), the optimization may choose to significantly increase feed rate on one furnace and significantly reduce feed on the other for very small benefit. Such a decision could easily lead to RTO credibility problems. In practice, these types of decisions can be avoided, e.g., by penalizing deviations from an average feed value.

Another modeling decision concerns the number of operating modes or equipment options on a given unit, such as the various combinations of heat exchangers that can be used for a given service. To allow for all combinations might significantly increase the complexity of the model. If some options are only used infrequently it may be better to simply not optimize for those cases. Note that not all relationships needed in the RTO come from first principle models. It is common, for example, to include key model relations from the MPC (e.g., hydraulic limits) in the RTO. Also, compressor efficiency curves are usually modeled based on a regressed equation.

A common practice following model development is to compare RTO model gains between independent and dependent variables with the corresponding gains available in the underlying MPC. The MPC gains are linear and derived from process step tests. The RTO model is generally considered valid if the RTO model gain and the MPC model gain match sign and order of magnitude in a similar range of operation (note: agreement within 20% or better is not uncommon for approximately linear effects). Exceptions to this rule may be made when dealing with severely non-linear models (e.g., valid models that may have gains that change sign). In those cases, engineering judgement is required to confirm the predictive ability of the RTO model.

5.3. Intermediate product prices

Several approaches are used in practice to determine prices of intermediate streams. Some simply take an open market price directly or discounted for the associated processing costs. A common approach is to add simple models to the RTO as a means to get an envelope large enough to capture feed and/or product prices, thereby avoiding the problem of intermediate prices. The planning group may also provide the price or price function based on their knowledge of marginal economics and refining LP results. Shadow prices from a planning LP may be appropriate to use in an RTO if yield changes or stream properties do not change significantly (for the expected range of RTO solutions); otherwise, the LP shadow costs should be corrected for deviations from base property values [21].

5.4. Model update

Selected model parameters are updated based on averaged plant measurements in order to keep the model in synch with the process. The model update parameterization problem is formulated as a simultaneous data reconciliation and parameter estimation problem (see e.g., [20]). Following the nomenclature used in [48], we define $\mathbf{z}$ as the vector of (noisy) measurements and $\mathbf{x}$ as the vector of model variables. Note that we do not distinguish between model inputs and outputs (i.e., independents and dependents). The vector $\mathbf{x}$ is partitioned into variables $\mathbf{x}^m$, which have a corresponding measurement value and $\mathbf{x}^u$, which do not and appear only in the model. The model parameters are expressed in terms of fixed values, $\boldsymbol{\alpha}$, and updated parameters, $\boldsymbol{\beta}$. Typically a weighted 2-norm objective function is used, which results in the following optimization problem:

$$\begin{aligned} &\min_{\mathbf{a},\boldsymbol{\beta},\mathbf{x}^u} \mathbf{a}^{\mathbf{T}}\mathbf{V}^{-1}\mathbf{a} \\ &\text{s.t.} \\ &\mathbf{f}(\mathbf{z}+\mathbf{a},\mathbf{x}^{\mathbf{u}},\boldsymbol{\alpha},\boldsymbol{\beta}) = 0 \end{aligned} \tag{3}$$

where $\mathbf{f}$ represents the vector of model equations, $\mathbf{a}$ the measurement adjustments or biases, and $\mathbf{V}$ is the covariance matrix of the measurement errors (typically diagonal). The design decision involves selecting which measurement adjustments and model parameters should be adjusted online automatically (i.e., $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$). This step is non-trivial and requires process experience and engineering judgement. The starting point is to consider which parameters/adjustments are affected by unmeasured disturbances and which subset of parameters/adjustments ensure a consistent model. Parameters that change on a slower time scale may not be appropriate to update every execution in order to minimize the influence of measurement noise and unmeasured disturbances.

There are differences among industrial practitioners regarding the number of measurements to use. Some choose to aim for a square problem (equal number of measurements and parameters/adjustments), or at least to limit the number of excess measurements. This is done to more readily allow subsections of the model to run in isolation and to facilitate troubleshooting of bad measurements. Others choose a larger number of measurements, selecting those which, via the model, interact with other measurements. For example, there may be 40 degrees of freedom in the parameterization case. A square problem would use 40 measurements, a non-square problem might use 100 measurements. Sometimes a non-square case cannot be avoided. A typical example is the estimation of crude unit feed composition based on crude assay information and product yields.

In practice, trial and error is used to select the measurements to use in the update, as a different subset of parameters/adjustments may lead to a better fit of the model vs. plant. Part of this pro-

Table 1
RTO and MPC example variables.

RTO variables		MPC variables	
Constraints	Decisions to MPC	Constraints	Manipulated setpoints (in DCS)
Reactor conversion	Desired targets	Temperature	Flow
Production rates	Min/max limits	Level	Temperature
MPC constraints	Costs/economic priorities	Composition	Pressure
		Column DP	valve positions
		Compressor power	
		Valve positions (PID outputs)	

cess is to look for significant jumps in the parameters/adjustments, which would be indicative of ill-conditioning due to a poor choice of update parameters/measurements or, once online, could arise due to a change in operating conditions or a missing measurement. The problem of detecting ill-conditioning in real-time has been studied and a solution proposed by Miletic and Marlin [31]; however, in practice the most that is typically done is to perform bound checks on updated parameters. Violations can be used to skip the optimization or flagged for later analysis. Some parameters, such as those related to catalyst deactivation, may require multiple data sets [47]. Current industrial practice is to perform these updates off-line.

5.5. Economic optimization

The current optimal economic operating point is determined by the optimizer using the plant model with the most recent updates of measurement adjustments, **a**, parameters **β**, constraint limits, and economic values. In most RTO applications, the objective function is profit (Product Value – Feed Costs – Variable Costs). The plant model includes material and energy balances, thermodynamic and kinetic expressions, and any additional relationships that may be required to model plant equipment or constraints. The optimization problem can be expressed as follows:

$$\begin{aligned}&\max_{\mathbf{x}} P(\mathbf{x}, \boldsymbol{\alpha}, \boldsymbol{\beta})\\ &\text{s.t.}\\ &\mathbf{f}(\mathbf{x}, \boldsymbol{\alpha}, \boldsymbol{\beta}) = 0\\ &\max(\mathbf{x}_{\mathrm{LB}}, \mathbf{x} - \Delta) \leq \mathbf{x} \leq \min(\mathbf{x}_{\mathrm{UB}}, \mathbf{x} + \Delta)\end{aligned} \quad (4)$$

P is the profit expression of the RTO. The bounds on the optimization variable **x** represents the minimum and maximum limits for the optimization variables; step limits, Δ, represents the maximum change, from the current value, that **x** is allowed to move in a single optimizer execution. Note that the optimal results for the model variables which are measured, is determined from the **a** of Eq. (3) as $\mathbf{z}_{opt} = \mathbf{x}_{opt}^{m} - \mathbf{a}$. Use of step limits can sometimes cause confusion with interpreting optimizer results; for example, an RTO variable may initially move in a direction opposite to where the RTO will ultimately go. Some implementations automatically solve an additional case to understand the decisions the optimizer would make without the step limitations.

5.6. MPC design issues

To integrate the RTO with the MPC, it may be necessary to add new controlled variables or manipulated variables to the MPC. The important design issues concern decisions as to which optimizer decisions should be passed to the MPC and which (if any) degrees of freedom should be left to the MPC. Here again, there is not a standard practice and it depends, in part, on the features and capability of the MPC. As mentioned in Section 3, desired RTO setpoints in the MPC can be implemented in various ways. These targets will be realized by the MPC, if they do not conflict with constraints in the controller. Typical setpoints include reactor conversion or severity. Another option is for the RTO to set a lower or upper limit of manipulated or controlled variables. Consider the case of a crude unit where the feed heater is duty constrained. The RTO could set a desired target for the feed rate, heater outlet, and product qualities. Since the optimizer may execute only every 2–4 h, it may be desirable to leave the feed as a degree of freedom in the MPC (to maximize in between RTO executions, if possible). In this case, it would make sense for the RTO to reset a lower heater outlet temperature. As a result, the MPC would not push feed rate to a level which would decrease the heater outlet below the RTO value (corresponding to more of the feed incrementally going out as lower valued resid product). In the case where degrees of freedom are left in the MPC, intrinsic variables are normally preferred (temperature, composition, ratio); in addition, MPC prices/costs must be properly determined so that the LP(QP) pushes to the correct set of RTO-determined limits.

An example of the types of variables that might be controlled and adjusted with RTO and MPC are given in Table 1. Note that the constraint variables in the RTO include as a subset the controlled and constraint variables of the MPC.

6. RTO challenges

Multiple challenges exist in successfully deploying and maintaining RTO applications. Some are of a technical nature. Others are a direct result of the decision hierarchy shown in Fig. 1, which requires coordination to ensure consistency and avoid conflict. Others are due to the increased demands that are placed on the entire organization.

Difficulties with RTO have been highlighted by Friedman [14] and Friedman [13], and include difficulty in determining prices for intermediate streams, feed composition estimation, coordination between MPC and RTO, challenge of updating steady-state models in a dynamic plant, and shortage of skilled staff to maintain the RTO. In addition to the wait time, Marlin and Hrymak [30] identified the following challenges: impact/coordination of RTO decisions on tanks and inventory, procedures for establishing model structure, accounting for uncertainty in models and data, and limitations associated with the use of typical operating data for updating parameter estimates.

Because RTO requires MPC, a properly performing MPC is a critical requirement. Gattu et al. [17] have identified the following characteristics of a properly performing MPC controller:

- The scope of the controller covers all key process constraints.
- All key equipment, process and product specifications constraints are controlled at limits.
- The economics push against the correct combination of constraints.
- Dynamic controller performance is good over a wide operating range.
- Intervention from the operator should only occur to handle rare conditions that were consciously not included in the controller design.

Even when the MPCs are performing well, the RTO can push operation to a different operating region and the MPC linear model may need to be updated or a linearizing transformation used. RTO also places a servo requirement on the MPCs, whereas most MPCs are tuned for regulatory control. Because both the MPC and RTO achieve economic objectives with different models, there can be challenges, particularly when degrees of freedom are left in the MPC. For example, in between RTO executions, an RTO-determined output target may not allow an MPC objective of, e.g., feed maximization, to be achieved. Also, if an MPC output (controlled variable) is targeted by the RTO, the MPC LP/QP may not (initially) accept it as feasible based on its linear model.

The most successful applications of RTO have been in ethylene plants. There are multiple reasons for this fact, including that the units often run at maximum production, there are complicated trade-offs that MPC cannot address, pricing and costs are well defined, and there is a lack of significant liquid inventories. Refining applications of RTO have been more problematic. The typical envelope (crude unit, FCC, etc.) for the RTO is often the same as for the MPC, which can limit the profit potential. Larger RTOs have been implemented, but these are the exception in the industry. Refinery units have the additional challenges that many of the products are intermediate ones and logistical constraints (from scheduling) often come into play. Practical challenges for crude and FCC unit RTOs are described by Hooper and Jones [23].

Good estimates of feed composition are critical to RTO as they impact the prediction of downstream flow rates. Most of the RTO projects require additional analyses, either lab and/or online measurements. A minimum requirement is a good estimate of the average composition corresponding to a particular feedstock. In general, measurements of more than bulk properties and a few components are required. For example, for an ethylene liquid feedstock, 30 components are typically required to characterize the feed.

An important issue that must be addressed in the modeling effort is the number of components to track through the process model. Due to the potentially large number of components in process streams (especially in refining), a reduced component slate or lumping of components is often used. Judicious selection of the components is required to minimize model errors. It is common that different components or lumps are used in the individual unit models (reactors vs. distillation), and translation of the components between subsections of the overall model is required [32].

Conflicts with the planning group are inevitable with an RTO, at least early in a project. These conflicts are minimized when the RTO is making decisions that the planning LP does not, for example, ethylene plant intermediate distillate column compositions, and refrigeration/column pressures; however, with other important variables such as cracking severities, both planning and the RTO try to optimize these. The potential conflicts in refining are greater as refining has additional complications such as gasoline pooling and blending, which interact with other units in the plant (and profit centers). Reactor models in the RTO can also lead to conflict with the simplified model in the planning LP. Unless convinced otherwise, the planning group/refining LP will probably have the "final word" in situations of conflict, possibly leading to the RTO not being used. However, if these conflicts are both expected and managed, then the coordination effort will lead to additional understanding and benefits.

Examples where consistency and coordination are required include the setting of economic values, identifying limiting plant constraints, constraint limits themselves, and model consistency. Notice that economics come into play at least at two levels – planning and RTO. Consistency of actual constraint limits is an important issue that affects planning, RTO, and MPC/operations. Another important coordination issue concerns the degrees and amount of freedom that are given to the RTO and MPC. Overly constrained freedom can lead to missed opportunities; too many or to much freedom can lead to problems with inventory and tankage, thus impacting the scheduling layer.

Model consistency is an important topic in its own right, but one that we will not discuss to any significant extent. Suffice it to say that consistency is required between planning and RTO, but the challenge lies in that the models are used to solve different business problems. In planning, the important issue is average plant behavior, important for decision related to feedstock purchases and operational planning "over time" (weeks and months). With RTO, the importance is the decisions that are made in "real time" based on current plant conditions rather than conditions assumed in the planning model.

With the benefit of hindsight, it is clear that the RTO system should not be viewed simply as the next level in the cascade or hierarchy, and that the effort and challenges in moving from MPC to RTO have been underestimated. Positional model accuracy requirements are higher for RTO than MPC, since the objective is steady-state optimization rather than dynamic regulation for MPC. This imposes a greater requirement for absolute accuracy of instrumentation. RTO also requires additional measurements compared to MPC, many of which are temperatures and pressures that historically may have not been regularly maintained. As a result, instrumentation issues often surface in an RTO project, which require instrument replacement or repair, particularly when the attempt is first made to reconcile the RTO model results to the plant data.

Given the challenges faced by industry in maintaining MPC applications, it is not surprising that RTO applications have been harder to maintain. RTO requires a higher level of expertise and commitment across the entire organization. This expertise has been hard to develop and maintain, especially for those companies without a strong central engineering group and technical career path. We should note that the specialized nature of the RTO engineer – detailed knowledge of control, modeling, optimization – is at odds with the desire for well-rounded generalist engineers and the typical career progression.

7. Alternative RTO formulations

In this section we present variations that have been taken in industrial applications to circumvent challenges associated with traditional RTO, and which we believe deserve additional investigation. A key contributor to the benefits of an RTO is the ability of the model to account for compositional affects that are implicitly linearized or averaged in the MPC models. As a result, the key factors driving alternative RTO formulations are achieving these compositional benefits in simpler formulations with less reliance on achieving steady-state operation.

As mentioned in the previous section, there are challenges to cascading RTO to MPC. Therefore, a natural question is whether RTO functionality can be moved to the MPC layer. One possibility is to use the nonlinear model to update the steady-state gains in the MPC, as shown in Fig. 3, instead of using the model in a separate optimizer. Thus a one-layer solution is obtained. With this approach, the MPC LP(QP) is used to determine the optimal operating point based on gains determined by the nonlinear model, without the requirement of achieving steady state (since in MPC the LP(QP) optimization is based on predicted steady-state values). By updating the gains, the shape of the economic surface is modified, and the LP(QP) performs a type of "hill climbing" optimization. This approach can find a smooth optimum through the use of bounds on the MV step sizes. Requirements for this approach include gain scheduling capability within the MPC and the use of true economics

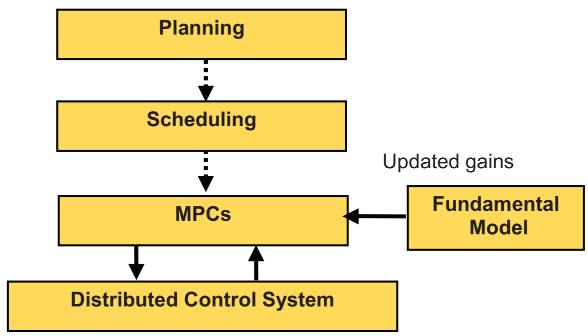

Fig. 3. Use of steady-state model to update MPC.

in the LP(QP). In addition, the nonlinear model needs to be updated to the plant. An FCCU application which updated the MPC gains based on a closed-form model of the FCCU is described by Jackson et al. [25]. Note that the model allows the tradeoff of FCC reactor riser outlet temperature and product yields. Another approach to a one-layer solution, also applied to an FCCU process, is given in Gouvea and Odloak [19], where the MPC control law was augmented with a nonlinear steady-state optimization based on MPC steady-state predictions at the end of the prediction horizon.

Alternative approaches utilizing an RTO–MPC cascade have been applied in ethylene plants. In these applications, feed maximizers are often configured with MPC. Without knowledge of furnace yield effects, a feed maximizer is unable to optimize furnace outlet temperature and simultaneously meet constraints on ethylene and propylene production. In Prior and Lopez [37] a simplified RTO for an ethylene plant was developed using a modular approach. A data reconciliation of a steady state mass and energy balance executed every 10 min, and the measurement offsets were filtered to compensate for dynamic effects. The model update was partitioned to allow individual sections of the plant to be updated asynchronously, based on steady state detection in the corresponding section of the plant. The optimization was scheduled on a 10 min interval using the latest parameter updates and the filtered biases from the data reconciliation. Optimizer outputs were passed to multiple MPCs. In Shindo et al. [40], the traditional RTO–MPC was used but only the cracking furnaces were modeled rigorously. Simple component splitters and mixers were used for the rest of the plant. In this application, interactions with the refrigeration systems were not modeled and there were no relationships to allow energy-separation trade-off in the distillation columns. The linear gains associated with constraints in the MPC were duplicated in the RTO. The use of MPC predicted steady state values instead of actual measurements allowed the RTO to execute at a much faster interval, nominally 30 min, which resulted in over 40 solutions/day being passed to a coordinator for the individual MPC controllers. In both of these examples, the component flows predicted by the furnace models are used to predict intermediate and final product flows.

Another approach that has been used industrially to remove the limitation of the steady-state wait time is based on simplified nonlinear dynamic models. These models are used as part of a traditional RTO cascade to individual MPCs. A grey-box modeling approach is taken where a static nonlinear model and an empirical dynamic model are used in a Hammerstein–Wieners structure, which is gain scheduled. In Arista et al. [1], the approach was applied across several refinery units to optimize the production of ultra low sulfur diesel. The non-linear model included mass balances, fractionation separation effects and non-linear property blending relationships. At each execution, the dynamic model predicts the steady-state of the process for use in the model update (parameterization) step, where the gains and other model parameters are adjusted to best fit the plant. A linear optimization is then performed.

A common aspect of these alternative formulations is they solve an important operational objective as opposed to providing a general purpose optimization. The emphasis is on identifying specific objectives for the system and designing a solution to meet those objectives.

8. Conclusion and suggested research directions

The advent of open equation modeling and SQP optimization techniques has enabled rigorous steady state optimizations to be formulated and reliably solved; however, sustained success across the industry has been spotty with real-time applications. Many applications are initially successful, but many companies have been unsuccessful in supporting and maintaining an RTO application after commissioning. As we have noted, RTO projects have generally been more successful in ethylene plant applications. While the challenges should not be minimized, we believe that the greater technical challenges associated with implementing RTO in refining and larger chemical plants can be largely overcome if there is sustained organizational commitment and access to engineering expertise. However, we note that in many refineries the lack of a modeling culture is an impediment to RTO. When there is an organizational commitment to RTO, the RTO model tends to be leveraged across the organization as an offline tool to help address issues such as understanding the impact of processing a new feed type, how to best manage a temporary plant reconfiguration (such as removing a heat exchanger from service due to a leak), or to provide more accurate yield information for the planning LP. While RTO may not be an attractive option for all companies, this does not take away the importance of rigorous models in offline use. As we have discussed, an offline model can be used to answer fundamental questions as to the potential benefits of optimization, the variability of the optimum and which independent variables have the greatest impact on profit, possibly leading to a policy that can be implemented more simply in MPC. A rigorous off-line model can also be used as the reference model for development and validation of a reduced model for online use. Or, more simply, a rigorous model could be used to determine the best set of controlled variables, i.e., those that result in minimal economic loss when controlled at constant setpoints, thereby avoiding the development of an online RTO model [43].

The steady-state wait time is a fundamental limitation of traditional RTO, regardless of the degree of rigor used. The wait time may or not be a significant issue for a given application. For example, a 2-h wait time may be acceptable for many refinery units, but bigger envelopes that arise from grouping refinery units or with larger petrochemical plants may be problematic. A larger optimization envelope is required for the optimizer to better address economic trade offs (e.g., a product pool such as gasoline) and to more directly manage logistical constraints. To address this issue in an RTO covering an Alkylation unit and FCCU, Georgiou et al. [18] provided the capability to update and optimize only one of the plants, if both are not at steady-state. An asynchronous model parameterization approach based on a partitioning of the plant (cf. Section 7) represents another option for addressing larger envelopes. An alternative formulation would be to keep the optimizers separate and use a coordination or integration strategy, possibly based on sharing of information among RTOs without simultaneous optimization of all variables in a single objective. While such an approach has not found widespread use with RTO, standard mathematical frameworks exist, including constraint and target based (primal), constraints/ targets plus economics, or price based (dual) [41]. As a case in point, consider the following simple example: two plants with separate RTOs produce the same product with an overall limit on the combined production rate. One way to coordinate these RTOs is to use a coordinating function to move the individual RTO production constraint limits until the shadow prices become equal. The general topic of coordinating individual RTOs warrants additional investigation. We note that a similar problem – the coordination of decentralized MPCs – is currently an active research area (see e.g., [45,4,29]).

As indicated in Section 7, linear dynamic models have also been used in practice to avoid the steady-state wait time. The basic idea is that predictions of steady state from a sufficiently accurate dynamic model (and updated based on the actual process values) would settle before the actual process settles at a new steady-state. This approach is worthy of investigation as it would retain the existing RTO structure: The subset of measurements selected for the model update/parameterization step would be replaced with their steady-state predictions. The model update/parameterization would then adjust these predictions (and model parameters) to arrive at a consistent steady-state. This approach might also be used as a means of bringing in feedforward information into RTO such as the prediction of ambient temperature. In the general case, the predictions would not have to be limited to just those dynamic models found in the MPC. A different model or modeling approach could be used. The dynamic models could be linear or nonlinear, with obvious implications of associated complexity.

A more fundamental approach to addressing the steady-state wait time issues would be to directly incorporate dynamic relationships. If the interest is steady-state optimization, as opposed to nonlinear control, the dynamic terms would impact only the model update case; the equations for the steady-state optimization would remain the same. With traditional RTO (with rigorous-based models), these additional equations, which result from the discretization of the corresponding differential equations, could prove to be problematic – i.e., in terms of solution time or complexity. Approximations/simplifications could be considered. For example, only the most significant dynamic effects and holdups could be modeled, in essence approximating the faster dynamic expressions with a steady state approximation. Also, empirical dynamic models mainly capturing the dominant time constants might be considered.

The use of hybrid or grey box models represents another possibility for introducing dynamics into the RTO model. By hybrid we mean combining experimental process data with known or approximated nonlinear knowledge or structure, such as with a Hammerstein–Wiener structure. Grey-box is used in the traditional sense of combining physical knowledge (conservations principles and rigorous constitutive equations) and empirically determined constitutive equations and parameter estimates. An approach that has been used industrially incorporates both dynamic process data and steady-state data from a fundamental model based on a gain-constrained neural-network model [39]. The goal of these approaches is to (at least implicitly) incorporate fundamental relationships to shape (constrain) model structure. An obvious requirement is one of having a sufficiently accurate description of the nonlinearity to track the true optimum closely. Following the point made under alternative formulations, the use of a hybrid or grey box approach probably necessitates that they be used to target a specific operational objective approach as opposed to performing a general purpose optimization; for example, addressing the problem of optimizing reactor conversion and downstream production. A hybrid or grey box approach, with its ability to incorporate nonlinear structure may offer an advantage over a single MPC or MPC coordinator (based on a common LP(QP), if it avoids the common problem of ill-conditioning often encountered with empirical models. An important issue in MPC applications is to ensure that the LP(QP) does not exploit fictitious degrees of freedom that may be introduced with an empirical model [8,22] for example, as a result of multiple controlled variables which essentially represent the same state.

As interesting research direction concerns the possibility of moving RTO functionality to MPC (and, possibly to the planning and scheduling layers). In the limit, this could lead to the RTO model becoming a reference model that is used to update the other models in the hierarchy. The previously cited approach of updating MPC gains based on a steady-state model represents the simplest case. We should note that simply updating certain model gains (or parameters) does not guarantee that the optimum reached by MPC will be the same as with the nonlinear model, because the NL model will have a different structure and more detail than the MPC model. However, since the nonlinear model is available, it would be possible to first evaluate a gain updating approach before deployment. An important issue to consider with this approach is its impact on the condition number of the MPC gain matrix, which could potentially introduce undesirable behavior (constraint vertex changes). Another possibility for moving RTO functionality to the MPC level is to provide MPC with additional knowledge of the optimal surface. The goal would be for the MPC to more profitably respond to disturbances that occur between RTO executions. In [24,3] an improved MPC target calculation is proposed to address this issue. The idea is to express the optimal MPC manipulated and controlled variables in terms of a 2nd order Taylor expansion in terms of key measured disturbance variables (e.g., feed composition and ambient temperature). Unmeasured disturbances might also be handled with such an approach. Some amount of additional complexity or nonlinear modeling might also be considered at the MPC level, but it should not lead to increased maintenance requirements or create convergence problems; the resulting optimization problem would need to solve (nearly) as reliably as linear MPC.

An issue that arises in all the RTO strategies covered here is model consistency in the MPC vs. those in the steady-state model, an important issue being gain consistency. The consistency requirements need to be better defined and techniques developed for analysis and enforcement. As discussed in Section 5, practitioners choose to try to match key MPC gains when developing the RTO model. However, this simple approach ignores the multivariable/directionality aspects of the model, which is arguably more important. We believe that analysis based on directions from an SVD or relative gain analysis (RGA) of MPC and RTO models would be a place to start.

Several of the alternative approaches incorporated simplified models. Simplified models may also have a role to play, even if a company has been successful with fully rigorous models. Examples include a simpler and more efficient online application or the development of a simpler model for another application (such as LP planning). Development of a simplified model can also be considered a problem of model reduction from a rigorous nonlinear static model to a reduced nonlinear static model. We assume that a nonlinear steady-state model is available to validate the approach, although the model may only be available in offline form (and all units may not be contained in the same model). A challenge is the lack of standard techniques for developing a simplified model. Another important issue is determining if a simplified model is appropriate for real-time optimization. This is not simply an issue of accuracy in predicting key variables. A requirement of the simplified model is that it converges to the Karush–Kuhn–Tucker (KKT) conditions of the rigorous model (1994). This implies that the simplified model contain a parameter set that can reproduce the optimum of rigorous model. Point-wise model adequacy requirements and criteria have been developed by [11]. Forbes and Marlin [10] developed a systematic approach to evaluating RTO design decisions termed design cost, which is defined as expected lost profit based on imperfections in the various design decisions, including model selection. The calibrated rigorous model can be used as the true plant to compare different modeling approaches. An important issue is how to develop or derive the simplified model. Standard techniques rely on compressing signals or mappings by expressing them as weighted sums of a small number of bases. The underlying techniques frequently rely on singular value decomposition or equivalent concepts. For such approaches to work satisfactorily it is important that process knowledge be included in the analysis. In practice these decisions are usually based on experienced modelers or trial and error. Simplified distillation models often incorporate mass and energy balances with a constitutive relationship (either empirical or grey box) to account for the separation effect (e.g., see [38]). Mahalec [28] has used such an approach to model refinery separation processes with the end goal of using the simplified model for both RTO and LP planning. Academic investigations into simplified approaches for common unit operations would be useful.

The importance of coordination and integration with the planning LP to avoid potential conflict has been discussed. Historically, RTO has been seen as the next level of the cascade to implement after MPC. A potentially better approach, after implementation of MPC, might be to work down from the planning LP, considering where model detail is lacking and considering what optimization functions need to be implemented in real-time. Further, using the planning LP to justify RTO, and to understand RTO's impact on plant wide margins, would help reduce the potential conflict. The following approach taken by [46] for a refinery hydrocracker RTO benefit estimation project is illustrative and may suggest ideas for further investigation. A version of the existing refining LP, without inventory transfers and using a fixed crude slate and product yields, was used to create stream prices for each of the past 12 months based on actual refinery feed and product prices. A nonlinear fundamental model of the hydrocracker was parameterized based on representative average data from each month and an optimization was performed for each data set. The nonlinear model exploited degrees of freedom that the refinery LP did not have (such as reactor bed temperatures and hydrogen flows). The LP was then updated based on a linearized model of the fundamental model to include the effects of the additional independent variables. The refinery LP showed similar results (direction and magnitude) and showed a similar increase on the refinery profit to RTO.

An area where more research attention should be given concerns which model parameters should be updated (and how often) and how to design experiments to more accurately estimate RTO model parameters. Poor selection of the update parameters can result in an ill-conditioned parameterization problem. The online use of multiple data sets, in a manner similar to moving horizon estimation, allows more parameters to be updated (particularly parameters changing slowly, such as catalyst activity). As previously noted, academic research in these areas has not found wide application in industrial applications. Design of experiments (DOE) is another area that has not found usage in RTO applications or been significantly invested by academia. Several RTO design problems associated with RTO can be formulated as a DOE. Here we mention a few that illustrate the usefulness of DOE techniques. Sensor selection for RTO is one and has been investigated by Fraleigh et al. [12]. An approach to online DOE has been suggested by Pfaff et al. [36], where excitation is added to the process as a means of reducing uncertainty of the optimization results, with constraints to minimize the expected profit loss from the experiments. A DOE approach for deciding on which model parameters should be selected for update has been formulated by Hahn, et al. [5] for the case of nonlinear dynamic models.

A final research area we wish to mention concerns the needs of end users of the RTO and personnel responsible for RTO maintenance. Improved diagnostic tools continue to show up on industrial wish lists. This includes tools to explain why the RTO reached the decision it did and how would the optimum change if a constraint bound is changed (this is also frequent request from MPC end users). On the maintenance side, tools or techniques to answer questions as to where and why an RTO model failed to converge are desirable.

Multiple challenges and technical issues have been raised. Ideally, these issues should be addressed jointly by industrial and academic investigators. Indeed, making meaningful contributions to the RTO area is a matter not merely of conceiving theoretically sound concepts, but also of incorporating technological, business, and people issues, as we have discussed. No single party can claim mastery of pertinent knowledge in such a diverse spectrum of issues. Moreover, most potential opportunities are based on sophisticated combination of many factors, such as robust and efficient computational tools, user-friendly and reliable software, skilled users, and organizational structures - hence the necessity for collaborative efforts. We believe that in recent years there has been insufficient dialog and collaboration between industry and academia with respect to RTO. Using history as a guide, it is fair to say that industrial initiative to promote MPC as the premier research topic among academicians, resulted both in improved theoretical understanding and better practical tools for MPC. In addition, it catalyzed the training of a number of engineers who are indispensable for making sophisticated industrial systems work. Similar developments for RTO would be welcome.

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